

# FreshTrack

Module Assignment & End User Documentation

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| Module Title      | <b>FreshTrack</b> — AI Food Freshness Detection Application                                     |
| Hosted URL (Live) | <a href="https://fresh-track-tau.vercel.app/">https://fresh-track-tau.vercel.app/</a>           |
| GitHub Repository | <a href="https://github.com/deepsh3969/FreshTrack">https://github.com/deepsh3969/FreshTrack</a> |
| Core Technologies | Google AI Studio (Vision API), Web-based Client                                                 |

## 1. System Overview

**FreshTrack** is an AI-powered food freshness detection application designed to actively mitigate food waste and enhance consumer health standards. Utilizing advanced visual recognition via Google AI Studio, the application analyzes uploaded or live-captured images of raw and prepared foods. It delivers instant freshness diagnostics, actionable consumption safety insights, vendor batch ratings, and smart culinary recommendations tailored to the item's exact state.

## 2. System Prerequisites

To access and seamlessly operate FreshTrack, users require the following minimal operational environment:

- **Web Browser:** Optimized for use on modern HTML5 browsers, primarily **Google Chrome** (Desktop or Mobile). Fully compatible with Microsoft Edge, Apple Safari, and Mozilla Firefox.
- **Hardware Access:** A device equipped with a built-in camera (for real-time photo verification) or access to local file storage to upload existing images.
- **Network Dependency:** A stable, active internet connection is mandatory to facilitate low-latency communication with the backend **Google AI Studio** vision processing models.

## 3. Getting Started Guide

Follow this step-by-step workflow to successfully log in and execute your first food quality verification:

1. **Navigate to Portal:** Open Google Chrome and access the live application hosted at <https://fresh-track-tau.vercel.app/>.
2. **Authenticate & Role Selection:** On the landing page, choose your respective operational persona. Log in as a **Customer** (for home pantry management) or as a **Vendor** (for inventory auditing and store ratings).

3. **Locate AI Scanner:** From the central dashboard or primary navigation panel, click on the prominent **Camera Icon** to open the AI Verification interface.
4. **Capture / Upload Food Item:** Select *"Upload Image"* to choose an existing photo from your file system, or select *"Take Photo"* to capture a live image. Ensure the target food item is clearly lit, centered, and free of obstructed packaging.
5. **Generate AI Analysis:** Submit the image. Within moments, the visual recognition engine will process the physical attributes of the item and render a complete diagnostic dashboard.

## 4. Key Features & How to Use

FreshTrack provides four core feature modules built to address the full lifecycle of food consumption and supply:

### **AI Food Freshness Verification**

**Functionality:** Evaluates external surface degradation, discoloration, wilting, and biological spoilage indicators in real-time.

**How to Use:** Following an image upload, look at the primary meter at the top of your diagnostic screen. The app calculates a definitive freshness score (e.g., *92% Optimal*, *Approaching Spoilage*, or *Unfit for Consumption*) alongside an estimated remaining shelf-life timeline.

### **Safety Insights & Pathogen Warnings**

**Functionality:** Assesses external health risks to prevent foodborne illness before consumption happens.

**How to Use:** Review the generated "Safety Report" section. The AI highlights spotted surface abnormalities (like early-stage mycelium/mold) and explicitly advises whether an item is safe for raw consumption or requires high-temperature cooking/boiling to eliminate surface pathogens.

### **Smart Recommendations & Zero-Waste Tips**

**Functionality:** Converts physical food status into practical, immediate culinary actions.

**How to Use:** Check the "Smart Actions" card. If an item is near its expiration threshold (e.g., overripe avocados or softening tomatoes), the system outputs immediate preservation methods—such as specific freezing instructions, pickling ratios, or targeted baking recipes.

### ★ Vendor Batch Ratings & Quality Badging

**Functionality:** A commercial feature allowing grocers and market vendors to maintain stock accountability.

**How to Use:** When logged in under the **Vendor** portal, users can sample-test newly arrived supply crates. Consistently high batch scores automatically grant the supplier a public-facing **"Verified Fresh Batch"** badge, boosting consumer trust across the platform.

## 5. Troubleshooting Common Issues

| Symptom / Error                       | Probable Root Cause                                                                      | Recommended Resolution                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>"Inconclusive Analysis" Notice</b> | Poor environmental lighting, extreme image blur, or multiple competing objects in frame. | Re-capture the subject under bright, neutral white light. Place the food on a plain, solid-colored background (e.g., a clean cutting board).                   |
| <b>Camera Failed to Initialize</b>    | The web browser has explicitly blocked hardware camera access for the domain.            | Click the padlock icon to the left of your URL address bar in Chrome, navigate to "Site Settings", and toggle Camera permissions to "Allow". Refresh the page. |
| <b>Prolonged AI Loading / Timeout</b> | High network latency or temporary Google AI Studio gateway congestion.                   | Verify your local Wi-Fi or cellular connection. If stable, wait 30 seconds and click the "Retry Analysis" button.                                              |